Project Documentation

CPL Chatbot System

ITC 6040 - Informatics Capstone

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December 2025

Table of Contents

Problem Statement

Credit for Prior Learning remains a proven mechanism for accelerating degree completion, particularly for working professionals and underserved populations. However, current CPL processes are predominantly manual, inconsistent, and resource-intensive. Students struggle to articulate professional experiences in academic frameworks, while faculty evaluators face overwhelming workloads reviewing unstructured submissions without standardized assessment tools. This creates significant barriers to participation and delays degree completion.

The chatbot will leverage IBM Cloud artificial intelligence to provide intelligent, conversational guidance through competency-based interviewing, structured evidence collection, and automated skills extraction. By integrating WatsonX.ai, WatsonX.Assistant, WatsonX.data, and WatsonX.governance, the system will maintain academic rigor while improving accessibility and consistency. This solution directly demonstrates how enterprise AI can address persistent higher education challenges, positioning IBM as a strategic partner in educational innovation.

I. Competitor Analysis

Purpose of Competitor Analysis

The primary objective of this competitive analysis is to evaluate the current landscape of AI chatbot solutions in higher education to inform our IBM CPL (Credit for Prior Learning) project development strategy. This analysis aims to:

- Identify market gaps and opportunities in the AI chatbot space specifically for Credit for Prior Learning applications
- Benchmark features and capabilities of existing solutions to establish competitive positioning
- Inform pricing strategy by understanding current market pricing models
- Guide product differentiation by identifying areas where IBM Cloud can provide unique value
- Assess integration requirements with existing university systems.

Business Background

Credit for Prior Learning (CPL) applications are complex, leading to information gaps, heavy consultant workloads, poor student experiences, and compliance issues. In the traditional model, students often struggle to match experiences or certificates with course outcomes, resulting in multiple submission rounds. Consultants face the time-consuming task of manually comparing resumes to course syllabi and providing repetitive feedback, which can lead to inconsistent judgments.

Chatbots can help by guiding students to identify missing information and providing instant feedback on their submissions, thus reducing the need for repeated submissions. They enhance transparency by linking responses to specific rubric sections and streamline consultant tasks, allowing them to focus on processing summaries rather than exhaustive reviews. This scalable approach improves clarity for students and reduces the burden on consultants.

Identify Competitors

Our analysis focuses on six key competitors in the AI chatbot market for higher education, including both direct and indirect competitors:

Direct Competitors (Higher Education-Focused):

- Ivy.ai - AI-driven chatbots specifically for higher education
- BlackBeltHelp - AI-powered helpdesk solutions for higher education
- CogAbility - AI-powered digital employees for government, healthcare, and education

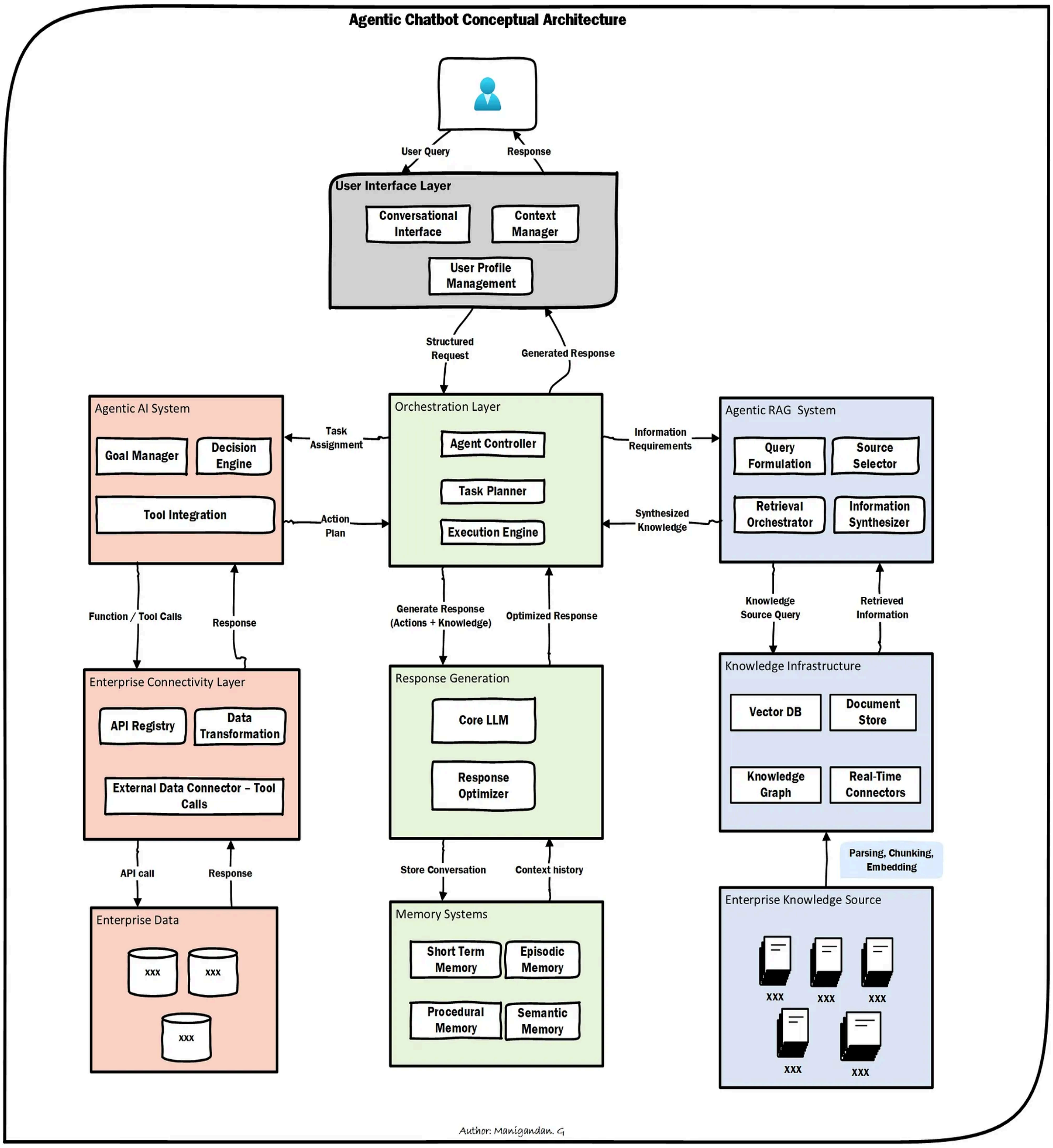


Fig: Chatbot Conceptual Architecture

Gather Key Data

Product/Service Offerings

Ivy.ai

Company Profile:

- Focus: AI-driven chatbots explicitly tailored for higher education institutions
- Core Business: Supporting the entire student lifecycle—from admissions and enrollment to advising and financial aid

- Key Features: Trained on millions of student inquiries, rapid deployment, prebuilt integrations with SIS, LMS, and CRM systems
- Primary Customers: Prominent colleges and universities
- Strengths: High scalability, advanced analytics, compliance with educational standards
- Limitations: Limited to education sector, little flexibility for non-academic workflows

BlackBeltHelp

Company Profile:

- Focus: AI-powered helpdesk solutions for higher education institutions and school districts
- Core Business: Unified communication channels and automated student support services
- Key Features: OneStop platform, 24/7 omnichannel support, SMS campaigns, helpdesk automation
- Primary Customers: Higher education institutions and school districts
- Strengths: Deep integration with student services, support for Credit for Prior Learning workflows
- Limitations: Narrower focus on academic support, less adaptable to broader organizational needs

Ada

Company Profile:

- Focus: No-code, scalable chatbot platform serving multiple industries
- Core Business: Automating customer and student support without technical expertise
- Key Features: Rapid deployment, multilingual capabilities, extensive channel support
- Primary Customers: SaaS, e-commerce, and education organizations
- Strengths: Highly flexible, easy implementation, versatile automation
- Limitations: Lacks built-in workflows for academic processes, requires custom development for CPL

CogAbility

Company Profile:

- Focus: AI-powered digital employees (CogBots) for public sector
- Core Business: Automating routine administrative tasks and conversational support
- Key Features: Pre-trained bots, document handling, easy implementation

- Primary Customers: Government, healthcare, and education sectors
- Strengths: Full-service deployment, retraining capabilities, affordability for public sector
- Limitations: Smaller market reach, less focused on student-specific workflows

Google Dialogflow

Company Profile:

- Focus: Natural language understanding platform for conversational interfaces
- Core Business: Intent recognition and context-aware interactions
- Key Features: Seamless integration with Google Cloud, multilingual capabilities, scalability
- Primary Customers: Various industries including education (Penn State, University of Birmingham)
- Strengths: Best-in-class NLU, extensive integrations, user-friendly design
- Limitations: Not education-specific, requires setup expertise

Microsoft Azure Bot Service

Company Profile:

- Focus: Intelligent conversational agents within Azure AI ecosystem
- Core Business: Creation, deployment, and management of chatbots across channels
- Key Features: Integration with Azure Cognitive Services, LUIS, Q&A Maker, Text Analytics
- Primary Customers: Universities and enterprises (IU International, University of South Florida)
- Strengths: Enterprise security, comprehensive AI ecosystem, Microsoft Teams integration
- Limitations: Setup complexity, not education-specific

Marketing and Revenue

Market Share and Growth:

- Ivy.ai and BlackBeltHelp dominate the higher education-specific market
- Ada holds significant market share in the broader SaaS chatbot space
- Google Dialogflow and Microsoft Azure have substantial enterprise market presence
- CogAbility maintains a smaller but focused public sector presence

Pricing Models:

- Ivy.ai: Premium pricing for specialized education features

- BlackBeltHelp: Competitive pricing for higher education institutions
- Ada: Flexible pricing tiers based on usage and features
- CogAbility: Cost-effective pricing for public sector organizations
- Dialogflow/Azure: Usage-based pricing models

Marketing Strategies:

- Education-focused vendors leverage academic conferences, partnerships with university vendors
- General platforms use broad digital marketing, enterprise sales teams, cloud marketplace presence
- All vendors emphasize case studies and success stories in their marketing

Customer Feedback

Common Strengths Across Competitors:

- Improved response times for student inquiries
- Reduced workload for administrative staff
- Enhanced 24/7 availability for student support
- Integration capabilities with existing systems

Common Pain Points:

- Setup complexity for non-technical users (particularly Dialogflow and Azure)
- Limited customization without technical expertise
- Generic responses that lack context for specific institutional needs
- Integration challenges with legacy university systems

Comparative Analysis

Competitive Profile Matrix (CPM)

Factor	Ivy.ai	BlackBeltHelp	Ada	CogAbility	Dialogflow	Azure Bot
Education Specificity	5	5	3	3	2	2
AI Capability	4	3	4	3	5	5
Integration	4	4	4	3	5	5
Customization	4	3	4	3	4	4
Scalability	4	4	5	3	5	5
Support/Service	4	3	3	2	3	4

Table 1: CPM, Scale: 5=Best-in-class; 1=Poor

Value Proposition Canvas (Summary)

Vendor	Gains (Benefits)	Pain Relievers	Products/Services
Ivy.ai	Fast answers, tailored to college/university	Trained on campus content, deep edu workflows	Chat, SMS, web, admin portal, integrations
BlackBeltHelp	24/7 instant support, reduced staff workload	Omnichannel approach, wide HE adoption	AI chat, phone, knowledge base, analytics
Ada	Handles huge volume, rapid deployment	Automation connects easily to CRM, LLMs	Web chat, CRM bots, self-service builder
CogAbility	Contextual, secure, accurate answers	Security-rich, compliance-driven design	AI chatbot, search, knowledge graph
Dialogflow	Multi-language, open platform, Google NLP	Easy integrations, enterprise security, cloud scale	Conversational AI, text/voice bot, analytics
Azure Bot Svc	Cross-campus omnichannel, enterprise AI	Microsoft ecosystem, NLP, compliance/global support	Teams, web, mobile, NLP, LUIS, analytics

Table 2: Value Proposition Canvas

SWOT Analysis

SWOT Analysis Matrix

Vendor	Strengths	Weaknesses	Opportunities	Threats
Ivy.ai	Education focus, robust AI, pre-trained KBs	US/EU focus, midsize player, costly customization	Expanding to more domains/AI integrations	Growing competition, integration demands
BlackBeltHelp	Large HE base, omnichannel, fast response	Can feel impersonal, less flexible with deep queries	Growth in voice, analytics, and integrations	Could face churn vs. agile LLM/NLP startups

Ada	Strong NLP, rapid automation, scalable	May lack education-native features, generic platform	Deeper sector customization, LLM partnerships	Intense SaaS/AI competition, evolving privacy rules
CogAbility	Advanced search/match, strong security claims	Smaller sector footprint, new to higher ed	Gov/edu sector expansion, compliance leverage	Risk from established HE-specific competitors
Google Dialogflow	Best-in-class NLU, many integrations, scalable	Not education-specific, requires setup expertise	Deep AI features, global edu adoption	Compliance sensitivity, indirect support channel
Microsoft Azure Bot	Enterprise security, AI ecosystem, MS Teams	Setup complexity, pricing, not education-specific	All-in-one campus bots, Azure AI integration	Privacy scrutiny, direct competition from Google/others

Table 3: SWOT Analysis Matrix

Market Positioning and Strategic Moves

Current Market Positioning

Education-Specific Leaders:

- Ivy.ai positions itself as the premium, comprehensive solution for higher education with deep vertical expertise
- BlackBeltHelp focuses on practical, omnichannel support with strong adoption in higher education

Platform Players:

- Ada positions as the accessible, no-code solution for organizations wanting quick deployment
- Google Dialogflow leverages Google's AI leadership and cloud ecosystem integration
- Microsoft Azure Bot Service capitalizes on Microsoft's enterprise relationships and Teams integration

Niche Player:

- CogAbility targets compliance-heavy, security-focused public sector organizations

Recent Strategic Moves

Industry Trends:

- Increased integration with LLMs and generative AI capabilities
- Focus on omnichannel experiences across mobile, web, and messaging platforms
- Enhanced analytics and reporting capabilities
- Emphasis on compliance and data privacy in educational settings

Notable Developments:

- Integration partnerships with major university system vendors
- Expansion of multilingual capabilities
- Development of industry-specific templates and workflows
- Enhanced security and compliance features for educational data

Insights and Recommendations

Key Market Opportunities for IBM CPL Project

Identified Market Gaps:

1. Specialized CPL Workflows: Current solutions lack specific Credit for Prior Learning process automation
2. Assessment Customization: Limited flexibility in configuring assessment criteria and rubrics
3. User Experience Gap: Most solutions require technical expertise for meaningful customization
4. Cost Accessibility: Enterprise-level features often priced out of reach for mid-size institutions

Strategic Recommendations

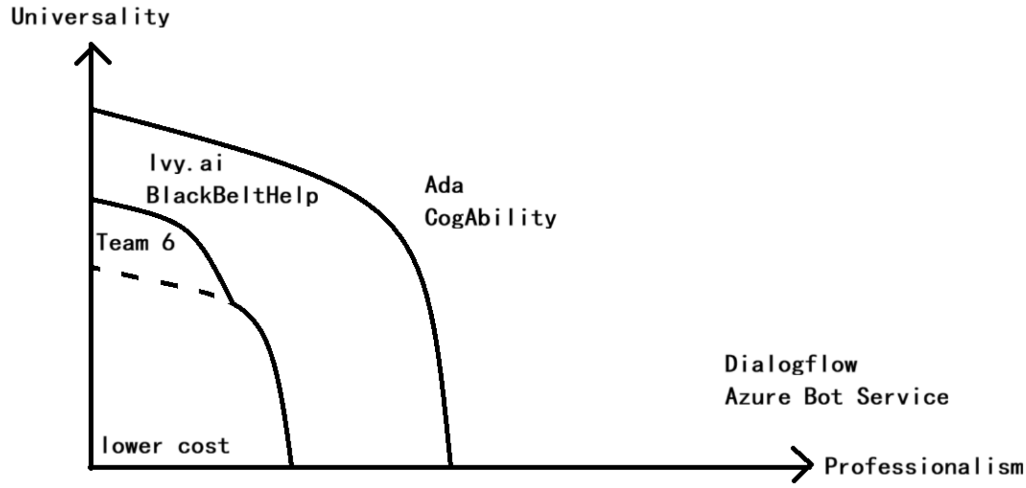


Fig 2: Strategic Graph

Competitive Positioning Strategy:

1. Open and Pluggable Assessment Criteria: Develop easily configurable business processes and review criteria through individual profiles
2. User-Friendly No-Code/Low-Code Tools: Balance intuitive visual builders with advanced coding capabilities
3. Persistent, Personalized User Memory: Maintain context across sessions without complex setup requirements
4. Agile Support and Community: Emphasize rapid support responses and active community engagement
5. Cost-Effective, Accessible Pricing: Flexible pricing tiers for educational institutions

Threats to Monitor

Competitive Threats:

- Rapid AI advancement by tech giants (Google, Microsoft)
- Potential entry of education-focused companies into the CPL space
- Price competition from established players
- Integration complexity with legacy university systems

Differentiation Strategy

Key Differentiators for IBM Solution:

- CPL-Specific Workflows: Purpose-built for Credit for Prior Learning processes
- Flexible Configuration: Easy customization without technical expertise

- IBM Cloud Integration: Leverage IBM's enterprise security and compliance capabilities
- Community-Driven Development: Open approach to feature development and customization
- Educational Pricing: Accessible pricing models for academic institutions

By combining IBM Cloud's core strengths with these differentiated offerings, Team 6 can target underserved markets and niche opportunities, gaining traction in the competitive landscape dominated by Ivy.ai and BlackBeltHelp. This positioning leverages IBM's strengths while addressing market gaps in agility, customization, and CPL-specific functionality.

II. Project Development Plan

Project Objectives

SMART Objectives

1. Functional Chatbot Delivery: Develop and deploy a production-ready voice and text-enabled chatbot prototype within a 10-week timeline using IBM Cloud technologies. Integrate competency-based interviewing methodology with skills extraction capabilities. Achieve 70-80% accuracy in evidence-to-competency mapping during pilot validation, with scalability potential through Docker containerization.

2. User Validation and Testing: Execute two rounds of pilot testing with 10-15 CPS Project Management students. Achieve minimum 80% user satisfaction rating on System Usability Scale. Validate that the system successfully handles 90% of expected interaction flows without critical errors. Document and implement user feedback for system refinement.

3. Governance and Compliance: Implement AI governance monitoring through WatsonX.governance to ensure ethical standards, performance transparency, and regulatory compliance. Establish model monitoring dashboards, performance baselines, and audit trails. Ensure 100% compliance documentation for academic and ethical standards.

4. Technical Documentation and Deliverables: Submit comprehensive design documentation, code repository, and live demonstration to IBM expert panel by week 14. Provide replicable architecture and deployment guides enabling future institutional adoption. Maintain architectural alignment with Northeastern University CPS standards and IBM Cloud best practices.

Sponsor Alignment

IBM seeks demonstrable use cases showcasing WatsonX's enterprise value in high-impact sectors, particularly higher education. This project directly addresses that strategic priority while solving a genuine institutional problem. Competitive analysis reveals that existing solutions (Ivy.ai, Dialogflow, Azure Bot, BlackBeltHelp) lack specialized CPL workflows. Our solution fills this market gap through domain-specific competency frameworks, CBI methodology integration, and embedded governance, directly showcasing WatsonX's advanced capabilities.

CPS - PJM requires automation reducing faculty workload while maintaining educational rigor. The chatbot delivers both operational efficiency and quality assurance through structured, consistent competency assessment aligned with institutional standards.

Project Scope

In Scope

- Design and develop voice-enabled AI chatbot using IBM Cloud (WatsonX.ai, WatsonX.Assistant, Speech-to-Text/Text-to-Speech services)
- Implement Retrieval-Augmented Generation (RAG) for context-aware guidance from CPL rubrics and competency frameworks
- Integrate competency-based interviewing methodology for structured evidence elicitation
- Develop dynamic assessment criteria with flexible feedback mechanisms
- Conduct pilot testing with CPS students and faculty evaluators
- Generate technical documentation, code repository, and live demonstrations
- Implement governance monitoring through WatsonX.governance

Out of Scope

- Integration with third-party portfolio management platforms (Blackboard, Desire2Learn)
- Real-time integration with NU CPS production systems or access to real student data
- Highly available, low-latency, cross-platform production deployment infrastructure
- Custom large language model training or fine-tuning (use of pre-trained WatsonX models only)
- Mobile native application development

Key Assumptions

- IBM will provide $500 WatsonX Cloud promotional credits covering development and testing

- CPS faculty will provide CPL sample applications, grading rubrics, and competency frameworks
- Development team has access to IBM Cloud documentation, SDKs, and sandbox environments
- User testing participants will provide informed consent and constructive feedback
- Project scope remains stable throughout the semester
- Team members complete IBM WatsonX certification training within first two weeks

Constraints

- Timeline: 10-week academic semester
- Resources: Three part-time student developers with concurrent coursework obligations
- Budget: $500 IBM Cloud credits plus minimal software licensing ($15/month estimated)
- Data Privacy: Legally prohibited from using real student data; testing uses synthetic/anonymized examples
- Infrastructure: Dependent on IBM Cloud service availability and API reliability
- Technical: No production CPS system access; limited to API-based integration

Team Members, Roles, and Responsibilities

Team Member	Role	Primary Responsibilities
Ummara Ali Syeda	Prompt Engineer & Database Developer (Team Lead)	Coordinate team meetings and project timeline; conduct user interviews; design and implement vector database for RAG pipeline; develop prompt engineering framework and intent taxonomy; build and optimize RAG pipeline; manage inter-service communication; conduct conversational flow quality assurance
Mengnan Wu	AI/Backend Developer	Analyze business requirements and NU IT DevOps environment; design system architecture and backend infrastructure; implement backend services using Spring Boot; integrate IBM WatsonX SDKs; develop data generators and warehouse; build CI/CD pipeline; design RESTful APIs; develop model trainers and RAG orchestration; conduct performance testing and monitoring

Shaoheng Zhang	Frontend Developer & UX Designer	Design wireframes and interactive prototypes; implement responsive web interface with voice and text modalities; conduct usability testing and iteration; ensure Northeastern CPS design compliance; implement WCAG 2.1 AA accessibility standards; create design system documentation and frontend code repository

Deliverables

Design Document and Technical Specification

Overview

Comprehensive technical and design specification serving as the authoritative reference for implementation, deployment, and ongoing maintenance.

Key Components

- System architecture diagrams with all IBM Cloud components and interactions
- Data flow diagrams illustrating user input through RAG pipeline, skills extraction, and feedback generation
- User interaction flows for voice and text modalities
- RAG model configuration including vector database schema, retrieval strategies, and ranking algorithms
- WatsonX service integration specifications with implementation details
- Competency-based interview question taxonomy and decision tree logic
- Security and privacy controls documentation
- Docker containerization strategy and scalability architecture
- Project development summary including training logs, test results, and lessons learned

Responsible Parties: Mengnan Wu (technical architecture), Shaoheng Zhang (UX/interaction design), Ummara Ali Syeda (RAG/prompt specifications)

Acceptance Criteria

- Faculty advisor review and approval
- Architecture diagrams professionally formatted with clear labels and captions
- All IBM Cloud service integrations documented with implementation specifications
- System supports voice and text interactions as specified
- RAG pipeline achieves 70% minimum accuracy in evidence matching
- Document contains 15+ pages of substantive technical content
- Deployment strategy clearly defined

Testing Plan

Internal review by faculty advisor; peer review by team members; validation against project requirements

Target Completion: End of Week 5

Proof-of-Concept Voice-Enabled Chatbot

Overview

Fully functional prototype demonstrating intelligent CPL portfolio assistance through conversational AI, evidence extraction, and real-time guidance.

Key Features

- Natural Language Interface: Multi-turn conversations using WatsonX.Assistant enabling human-like dialogue
- Dual-Mode Interaction: Voice input (IBM Cloud Speech-to-Text) and traditional text-based chat
- Skills Extraction: NLP-powered system identifying competencies and matching against CPL rubrics
- Competency-Based Interview Engine: Structured questioning following CBI methodology
- Dynamic Feedback: Real-time guidance on evidence quality and portfolio improvement
- Context-Aware Responses: RAG-powered retrieval of relevant CPL policies and example portfolios
- Session Management: Save, resume, and export interview sessions with structured portfolio drafts
- Responsive Design: Mobile-optimized interface aligned with Northeastern CPS standards
- Accessibility Compliance: WCAG 2.1 AA compliance with keyboard navigation, screen reader support, multi-language capability

Responsible Parties: Mengnan Wu (backend, AI integration), Shaoheng Zhang (frontend, UI implementation), Ummara Ali Syeda (prompt engineering, RAG optimization)

Acceptance Criteria

- Chatbot operates in IBM Cloud environment without critical errors
- Voice processing achieves max accuracy baseline (IBM Cloud Speech-to-Text standard)
- Text responses generated within 2-3 seconds

- Skills extraction achieves 70-80% accuracy against faculty baseline assessments
- Supports minimum 1 distinct CPL competency domains
- Successfully completes 90%+ of predefined interaction flows
- Secure session data storage with encryption
- Generates exportable portfolio summaries with structured output
- Code contains inline documentation and technical specifications

Testing Plan

Unit Testing: Function-level testing for NLP, database queries, API calls; 85%+ code coverage using JUnit and Jest

Integration Testing: WatsonX.Assistant, RAG pipeline, and backend service interactions; API communication verification with mock and live services

Usability Testing: Two pilot rounds with 10-15 CPS students; System Usability Scale ≥75; task success rate ≥80%

Performance Testing: Concurrent user load testing; response time benchmarking (<3 seconds); memory and CPU utilization monitoring

Target Completion: End of Week 8 (prototype); refinements through Week 11

AI Governance and Monitoring Setup

Overview

Comprehensive governance protocols through WatsonX.governance ensuring ethical AI practices, regulatory compliance, and transparent performance tracking.

Key Components

- Real-time model monitoring dashboard tracking accuracy, latency, and satisfaction metrics
- Automated bias detection for fairness assessment across competency domains
- Data usage compliance documentation and FERPA alignment
- Performance metrics tracking (accuracy, precision, recall, F1 scores)
- Complete audit trail of model decisions, user interactions, and system changes
- Automated alerts when performance metrics deviate from baselines
- Model explainability documentation for decision transparency
- Quarterly assessment reports with performance analysis and recommendations

Responsible Parties: Mengnan Wu (infrastructure), Ummara Ali Syeda (governance configuration)

Acceptance Criteria

- WatsonX.governance dashboard fully operational
- All model performance metrics logged and retrievable
- Bias detection mechanisms active with anomaly alerts
- Complete audit trail maintaining 100% interaction records
- Weekly compliance reports generated with zero critical violations
- Governance framework documentation explaining ethical AI standards compliance
- Model explainability features enabling understanding of 95%+ of system decisions

Testing Plan

Configuration validation within WatsonX.governance; monitoring accuracy verification with synthetic data; compliance audit with sample interactions

Target Completion: End of Week 9

Code Samples Repository

Overview

Production-quality GitHub repository containing complete implementation code, deployment configurations, and comprehensive documentation.

Key Components

- WatsonX.Assistant integration code (skills, intents, entities, conversation flows)
- RAG pipeline implementation (vector database operations, retrieval algorithms)
- Spring Boot backend microservices (API management, session handling, data processing)
- React/Vue frontend components (chatbot UI, voice controls, accessibility features)
- API documentation (OpenAPI/Swagger specifications with examples)
- Database schemas (SQL and vector database configuration)
- CI/CD configuration (Docker files, GitHub Actions workflows)
- Environment configuration (sample .env files, configuration management)
- Testing suites (unit tests, integration tests, user testing scripts)
- Deployment guide (step-by-step instructions for IBM Cloud environments)
- Comprehensive README with setup, usage, and troubleshooting

Responsible Parties: All team members (respective code ownership); Mengnan Wu (repository management)

Acceptance Criteria

- Repository accessible to IBM and Northeastern evaluators
- Consistent naming conventions and style guidelines documented
- 80%+ of functions contain explanatory code comments
- Commit history demonstrates regular contributions from all team members
- README includes executable setup and deployment instructions
- Complete API documentation with 10+ example requests/responses
- CI/CD pipeline automatically builds, tests, and reports quality metrics
- No hard-coded credentials or sensitive information in repository
- License file included (MIT or Apache 2.0)
- Minimum 50 commits demonstrating iterative development

Testing Plan

Code review by team members; successful builds verified; deployment instructions validated on clean IBM Cloud environment; automated security scanning

Target Completion: Core repository by Week 10; final code by Week 13

Final Presentation and Live Demonstration

Overview

Comprehensive presentation to IBM experts and Northeastern University faculty demonstrating chatbot capabilities, technical implementation, and business value.

Key Components

- Executive summary (5 minutes): Problem, solution, business impact
- Live demonstration (10 minutes): Interactive chatbot in action
- Technical deep dive (10 minutes): Architecture, IBM Cloud integration, RAG, governance
- User testing results (5 minutes): Pilot outcomes, quantitative metrics, user feedback
- Business case (5 minutes): Scalability, ROI analysis, deployment pathway
- Lessons learned (3 minutes): Technical challenges, solutions, future recommendations
- Q&A session (5-10 minutes): Expert questions and discussion
- Supporting materials: 15-20 professional slides, demonstration script, technical handouts

Responsible Parties: All team members (collaborative); Ummara Ali Syeda (overall lead); Mengnan Wu (technical section); Shaoheng Zhang (UX/testing section)

Acceptance Criteria

- Presentation completed within 30-minute time slot
- Live demonstration executes without critical errors
- All team members present and competently answer questions
- Professional slide formatting with consistent branding
- Accurate technical content appropriately pitched to expert audience
- User testing data clearly presented with visualizations
- All deliverables successfully demonstrated
- Presentation addresses IBM and CPS strategic objectives alignment

Testing Plan

Internal dress rehearsal with faculty advisor; timed run-through; contingency plans for technical failures

Target Completion: Presentation prepared by Week 13; delivery in Week 14

Schedule and Timelines

Work Breakdown Structure

Phase 1: Initiation and Planning (Weeks 1-2)

Task	Responsible	Dependencies	Deliverables
Project Kickoff Meeting	Ummara Ali Syeda	None	Meeting notes, stakeholder agreement
Team Onboarding & IBM WatsonX Training	All	Kickoff	Training certificates, environment access
Requirements Analysis	Ummara Ali Syeda, Mengnan Wu	Training	Requirements specification
Competitive Analysis	Ummara Ali Syeda	None	Competitor comparison matrix
CPL Process Mapping	Ummara Ali Syeda, Faculty	Kickoff	Process flows, rubric documentation

Preliminary Architecture Review	Mengnan Wu, Faculty	Requirements	Architecture draft

Milestone: Project Charter approved; team trained and ready for design

Phase 2: Design and Planning (Weeks 3-5)

Task	Responsible	Dependencies	Deliverables
Detailed System Architecture Design	Mengnan Wu	Phase 1	Architecture diagrams, specifications
RAG Pipeline Design	Ummara Ali Syeda, Mengnan Wu	Architecture	RAG workflows, vector DB schema
Chatbot Conversation Flow Design	Ummara Ali Syeda	Architecture	Intent taxonomy, conversation trees, prompts
Database Schema Design	Mengnan Wu	Architecture	SQL schemas, vector DB configuration
UI/UX Wireframing and Prototyping	Shaoheng Zhang	Architecture	Wireframes, interactive prototypes
API Contract Definition	Mengnan Wu	Architecture	OpenAPI/Swagger specifications
Accessibility and Compliance Planning	Shaoheng Zhang	Architecture	WCAG 2.1 AA checklist
Security and Privacy Design	Mengnan Wu	Architecture	Threat model, privacy controls
Design Document Integration	All	All design tasks	Final design document (15+ pages)

Milestone: Comprehensive Design Document approved by faculty and IBM

Phase 3: Development and Implementation (Weeks 6-8)

Task	Responsible	Dependencies	Deliverables
IBM Cloud Infrastructure Setup	Mengnan Wu	Phase 2	Cloud environment, CI/CD pipeline
WatsonX.Assistant Configuration	Ummara Ali Syeda, Mengnan Wu	Architecture	Intents, entities, skills defined
Vector Database Implementation	Mengnan Wu, Ummara Ali Syeda	Schema	Vector DB operational, populated
RAG Pipeline Development	Ummara Ali Syeda, Mengnan Wu	Infrastructure	RAG code, retrieval testing
Backend API Development	Mengnan Wu	Architecture	RESTful APIs implemented
Prompt Engineering and Optimization	Ummara Ali Syeda	WatsonX setup	Final prompts, intent taxonomy
Frontend Development	Shaoheng Zhang	UX design	React/Vue components, responsive design
Voice Integration	Mengnan Wu, Shaoheng Zhang	Frontend	Speech-to-Text, Text-to-Speech
Session Management Implementation	Mengnan Wu	Backend	Session storage, portfolio export
Security Implementation	Mengnan Wu	Architecture	Authentication, encryption, access control
Accessibility Features Implementation	Shaoheng Zhang	Frontend	WCAG 2.1 AA, multi-language support
Integrated System Testing	Mengnan Wu, Ummara Ali Syeda	Core development	Integration tests, bug reports

Milestone: Functional PoC chatbot operational in IBM Cloud; core features working end-to-end

Phase 4: Testing and Validation (Weeks 9-11)

Task	Responsible	Dependencies	Deliverables
Unit Testing	Mengnan Wu, Shaoheng Zhang	Development	Unit test reports, code coverage
Integration Testing	Mengnan Wu, Ummara Ali Syeda	Phase 3	Integration test results, metrics
WatsonX.Governance Setup	Mengnan Wu, Ummara Ali Syeda	System deployment	Governance dashboard, monitoring
Model Accuracy Validation	Ummara Ali Syeda, Mengnan Wu	Governance	Accuracy results, precision/recall
Performance and Load Testing	Mengnan Wu	System operational	Performance reports, scalability
Accessibility Compliance Testing	Shaoheng Zhang, Mengnan Wu	Frontend	WCAG 2.1 AA report, audit results
User Pilot Testing - Round 1	Shaoheng Zhang, Ummara Ali Syeda	PoC	User feedback, SUS scores, task metrics
Feedback Analysis and Iteration	All	Round 1	Bug fixes, UX improvements, updates
User Pilot Testing - Round 2	Shaoheng Zhang, Ummara Ali Syeda	Phase 4.8	Final feedback, satisfaction metrics
System Refinement	All	All testing	Updated code, bug fixes, documentation

Milestone: User satisfaction achieved (≥80%); accuracy targets met (70-80%); system ready for demonstration

Phase 5: Delivery and Demonstration (Weeks 12-14)

Task	Responsible	Dependencies	Deliverables
Code Repository Finalization	Mengnan Wu, All	Development	Final repository, documentation
Technical Documentation Completion	Mengnan Wu, Ummara Ali Syeda	Code finalization	API docs, deployment guide
Presentation Development	All	Testing	Slides, demo script, materials
Final Presentation Rehearsal	All	Presentation	Polished delivery, timing verified
Demonstration Materials Preparation	All	PoC	Demo scenarios, backup recordings
Project Documentation Consolidation	All	All phases	Lessons learned, recommendations
Final Presentation and Delivery	All	All deliverables	Live demonstration to IBM and faculty
Post-Presentation Documentation	Ummara Ali Syeda	Presentation	Feedback summary, follow-up docs

Milestone: All deliverables completed and accepted; successful presentation delivery

Project Schedule

Total Duration: 14 Calendar Weeks (10 Academic Weeks)

Week	Initiation	Design	Development	Testing	Delivery
1	Kickoff, Training				
2	Requirements				
3		Architecture Design			

4		Design Phase			
5		Design Complete			
6			Infrastructure, RAG		
7			Core Development		
8			PoC Functional		
9				Unit/Integration Testing	
10				Governance Setup	
11				User Testing Complete	
12					Documentation
13					Presentation Prep
14					Final Presentation

Key Milestones

Milestone	Target Week	Success Criteria
Requirements Complete	Week 2	Signed requirements document, no critical gaps
Design Document Approved	Week 5	15+ pages, faculty approval, architecture validated
Functional Prototype Ready	Week 8	90%+ interaction flows successful, operating in IBM Cloud

User Testing Complete	Week 11	≥80% satisfaction, 70-80% accuracy achieved
All Deliverables Ready	Week 13	Repository complete, documentation finalized
Final Presentation	Week 14	Successful demonstration, deliverables accepted

Planning Documents

Resource and Budget Plan

Personnel Resources

Resource	Role	Allocation	Cost
Shaoheng Zhang	Frontend Developer & UX Designer	15 hours/week × 14 weeks	In-kind (student labor)
Mengnan Wu	AI/Backend Developer	15 hours/week × 14 weeks	In-kind (student labor)
Ummara Ali Syeda	Prompt Engineer/Database Developer	15 hours/week × 14 weeks	In-kind (student labor)
Faculty Advisor	Project oversight and guidance	~2 hours/week × 14 weeks	In-kind (faculty time)
IBM Liaison	Technical support and resources	~1 hour/week × 14 weeks	In-kind (IBM support)

Software and Services

Resource	Purpose	Cost	Coverage
IBM WatsonX Cloud Services	Core development platform	$0 ($500 promo code)	Speech-to-Text, NLP, vector database, governance
IBM Cloud Storage	Session data, logs, database	Included	Within $500 allocation
GitHub Repository	Version control, CI/CD	$0 (free tier)	Unlimited public/private repos

Figma Professional	UI/UX design, prototyping	$12/month × 4 months	Collaborative design, components
Google Workspace	Documentation, collaboration	$0 (NU account)	Docs, Sheets, Slides
Docker Hub	Container registry	$0 (free tier)	Image storage
Development Tools	IDEs, testing frameworks	$0 (open source)	VS Code, JUnit, Jest

Total Estimated Budget: $50-60 (software only; IBM credits provided by sponsor)

Budget Allocation

- 70% ($35-40): IBM WatsonX services
- 20% ($10-12): Design tools (Figma)
- 10% ($5-8): Contingency and miscellaneous

Risk Management Plan

Risk ID	Risk Description	Category	Probability	Impact	Score	Mitigation Strategy	Contingency
R1	IBM Cloud infrastructure unavailability during development	Technical	30%	High	6	Weekly monitoring; maintain a local code backup environment; mock services for development	Switch to local deployment; negotiate timeline extension
R2	Inadequate WatsonX training affecting architecture	Technical	35%	High	7	40-hour comprehensive training; weekly IBM office hours; expert architecture review	Engage external IBM consultant for review

R3	User testing reveals critical usability issues late (Week 11+)	Technical	40%	High	8	Early validation with low-fidelity prototypes (Week 3-4); heuristic evaluation (Week 6); aggressive iteration window (Weeks 10-11)	1-week timeline extension if necessary
R4	Scope creep from sponsor mid-project	Organizational	35%	Medium	5	Define clear scope boundaries Week 2; formal change control process; bi-weekly scope reviews; written documentation	Prioritized backlog; implement only highest-priority features
R5	Team member unexpected unavailability	Organizational	25%	High	6	Detailed documentation; cross-training; pair programming; buddy system; written decision records	Redistribute workload; negotiate extension; backup resources
R6	Skills extraction accuracy falls below 70% target	Technical	40%	Medium	6	CPL assessment tools literature review; multi-strategy RAG pipeline; weekly monitoring; early prompt	Adjust target to 60%; document learnings for future

						optimization (Week 5-6)	
R7	Design complexity leads to inconsistent UX	Technical	25%	Medium	4	IBM Carbon Design System; single design system; component library (Week 3); phase-end design reviews; automated accessibility checks	Simplify feature set; extend design phase 1 week
R8	Data privacy and regulatory concerns	Organizational	20%	High	6	Engage NU Legal/Privacy Office Week 1; synthetic data exclusively; WatsonX.governance from start; document all decisions	Legal review and approval before pilot testing
R9	Third-party API integration issues	Technical	20%	Medium	4	Begin API testing Week 6; version control API changes; fallback paths; mock non-critical APIs	Identify alternative services; rapid implementation
R10	Insufficient user pilot participants (target: 10-15)	Organizational	25%	Medium	4	Early recruitment (Week 5-6); offer incentives; convenient scheduling;	Accept lower sample size (minimum 5); extend testing

						backup participant list	

Quality Management Plan

Quality Standards and Objectives

Quality Dimension	Standard	Target	Measurement
Functional Completeness	All features implemented and operational	100% scope coverage	Feature completion checklist, QA testing
Performance	Response time	<3 seconds for 95% of queries	Automated monitoring, load testing
Reliability	System uptime	99% during testing	Infrastructure logs
Accuracy	Skills extraction accuracy	70-80%	Validation against faculty assessments
Usability	User satisfaction	SUS ≥75, task success ≥80%	User testing, System Usability Scale
Security	Vulnerability status	Zero critical vulnerabilities	Security scanning, penetration testing
Documentation	Code documentation	≥80% functions documented	Code review
Maintainability	Code quality	Modular architecture	Code review, complexity metrics
Compliance	Governance and ethics	100% compliance documented	WatsonX.governance monitoring

Quality Assurance Processes

Design Phase:

- Peer review of architecture by external expert
- Requirements traceability matrix validation
- Northeastern CPS standards compliance verification

- Faculty feedback on CPL alignment

Development Phase:

- Code review for 100% of pull requests (minimum 2 reviewers)
- Automated static code analysis (SonarQube)
- Component integration testing before system integration
- Continuous integration pipeline on each commit
- Performance profiling and optimization

Testing Phase:

- Unit testing (85%+ coverage)
- Integration testing of major components
- Load testing with concurrent user simulation
- Usability testing (2 rounds, target 10-15 users)
- Accessibility compliance testing (automated + manual)
- Performance benchmarking
- Security penetration testing

Acceptance and Documentation:

- Defined, measurable acceptance criteria for each deliverable
- Faculty/IBM sign-off on criteria before development
- Verification testing confirming criteria met
- Documented evidence (test reports, logs, screenshots)

Quality Metrics and Review

- Weekly quality dashboard: Coverage, bugs, performance, test results
- Monthly quality review: Trend analysis, improvements, risks
- Milestone sign-off: Faculty and IBM approval of quality standards

Communication Plan

Stakeholder Communication

Stakeholder	Frequency	Channels	Content	Owner
Internal Team	Daily + Weekly	Teams, WhatsApp, Monday meetings	Task updates, blockers, progress	Team Lead, Ummara Ali Syeda

Faculty Advisor	Weekly + As-needed	Email, bi-weekly meetings, Teams	Major decisions, risks, milestones	All
IBM Liaison	Weekly + Urgent	Email, monthly calls, Teams	Technical questions, architecture, resources	Professor Zhou
CPS Stakeholders	Testing	Email, scheduled meetings	Requirements, testing, feedback	Project Managers
Project Managers	Weekly + Internal PMs weekly	Email, Teams.	Timeline, Overview, stakeholder communication	Vismit

Communication Channels

Channel	Purpose	Format	Frequency
Teams, Email	Quick updates, questions, coordination	Messages, threads	Daily
GitHub Issues	Task tracking, technical issues	Issues, pull requests, boards	Continuous
Weekly Team Meeting	Progress review, planning, issues	60 min, Mondays 10am	Weekly
weekly Faculty Meeting	Milestone review, guidance	30 min, Wednesdays 12 pm	Every week
Monthly IBM Sync	Technical consultation	60 min, Fridays	Weekly
Figma	Design collaboration	Comments, version history	During design phase
Project Wiki	Central documentation	Collaborative document	Weekly updates

Meeting Agendas

Weekly Team Meeting (60 min):

- Status updates (15 min)
- Blockers and escalations (10 min)
- Technical decisions (15 min)
- Risk review (5 min)
- Next week planning (15 min)

Bi-weekly Faculty Meeting (2 hours):

- Milestone progress (30 min)
- Deliverables review (15 min)
- Issues and guidance (45 min)
- Feedback (30 min)

Weekly IBM Sync:

- Architecture alignment (15 min)
- Technical challenges (60 min)
- Credits and resources (5 min)
- Timeline review (15 min)
- Q&A (60 min)

Feedback and Change Management

1. Feedback Collection: Weekly retrospectives (internal); bi-weekly faculty meetings (sponsor); user testing sessions (end-users)

2. Feedback Triage: Categorize (bug/enhancement/change request); assess impact/effort; prioritize (must-have/nice-to-have/future)

3. Change Request Process: Submit to team lead; assess scope/schedule/budget impact; review with faculty/IBM if significant; document decision; update project plan; communicate to stakeholders

4. Issue Escalation:

    - Critical issues: Escalate to faculty within 24 hours
    - High-priority risks: Discuss in next team meeting
    - Schedule/budget impacts: Require faculty approval

Documentation and Knowledge Management

- Project Wiki/Notion: Decisions, meeting notes, documentation
- GitHub Wiki: Technical documentation, architecture rationale, code standards
- Design System: UI/UX patterns, accessibility standards, implementation guides
- API Documentation: OpenAPI specs, examples, troubleshooting

- Deployment Guide: Setup instructions, configuration, troubleshooting

III. User Testing Plan

Outline

User testing plan outlines the evaluation methodology for the Credit for Prior Learning (CPL) Intelligent Assistant, an AI-powered conversational system designed to streamline the CPL process for students and faculty at Northeastern University. The testing plan aims to evaluate usability, effectiveness, and user satisfaction through a systematic approach involving 8-10 participants between 1 to 4 week.

The plan incorporates established usability testing methodologies (Nielsen, 1994; Villamañe, 2024) with educational technology-specific considerations. Eight comprehensive test scenarios cover all major system functionalities, from initial user orientation through complex multi-course waiver requests and AI-powered analysis interpretation.

Key testing objectives include evaluating ease of navigation, clarity of instructions, feature effectiveness, system comprehension, and error recovery capabilities. The hybrid testing approach accommodates diverse participant needs while ensuring comprehensive data collection through quantitative metrics and qualitative insights.

Expected outcomes include achieving 80% user satisfaction ratings, 90% task completion rates, and actionable recommendations for system improvement prior to deployment. The testing results will directly inform product development iterations and ensure the system meets user needs and academic standards.

Baseline Information

Objectives

This user testing plan aims to comprehensively evaluate the usability, effectiveness, and user satisfaction of the Credit for Prior Learning (CPL) Intelligent Assistant chatbot system. The testing objectives are strategically aligned with our current development stage and project goals of creating an AI-powered conversational interface that streamlines the CPL process while maintaining academic rigor and user confidence.

Primary Testing Objectives

Objective Category	Description	Success Metrics
Ease of Navigation	Evaluate how easily users can navigate through the chatbot interface, access different features, and complete the CPL submission workflow without assistance	Task completion rate max, Average time per task less, User satisfaction with navigation maximum
Clarity of Instructions	Assess whether users understand the chatbot's guidance, prompts, and feedback throughout the CPL process, including document upload requirements and evaluation criteria	Instruction clarity rating maximum, User comprehension accuracy maximum, Reduced help-seeking behavior
Feature Effectiveness	Determine the effectiveness of core features including document upload, AI-powered analysis, similarity scoring interpretation, and recommendation generation	Feature utilization rate max, User satisfaction with features >80%, Successful feature completion rate maximum

Objective Category	Description	Success Metrics
System Trust & Comprehension	Measure users' understanding of the CPL process as explained by the chatbot, including their confidence in the system's assessment capabilities and AI transparency	User trust score max, AI explanation comprehension max, Confidence in system decisions maximum
Error Recovery	Evaluate users' ability to recover from errors, handle edge cases, and successfully complete tasks despite encountering problems or system limitations	Error recovery success rate max, Error message clarity max, Time to resolve errors in less time

These objectives directly connect to our project goals of achieving maximum accuracy in evidence-to-competency mapping, user satisfaction rating, and successful handling of expected interaction flows, as established by recent research in educational AI systems (Chen et al., 2024; Villamañe, 2024).

Target Users

Primary User Groups and Personas

The CPL Intelligent Assistant targets three primary user personas, each with distinct needs, technological proficiency levels, and usage contexts. Understanding these user groups is essential for designing relevant test scenarios and interpreting results within appropriate contexts.

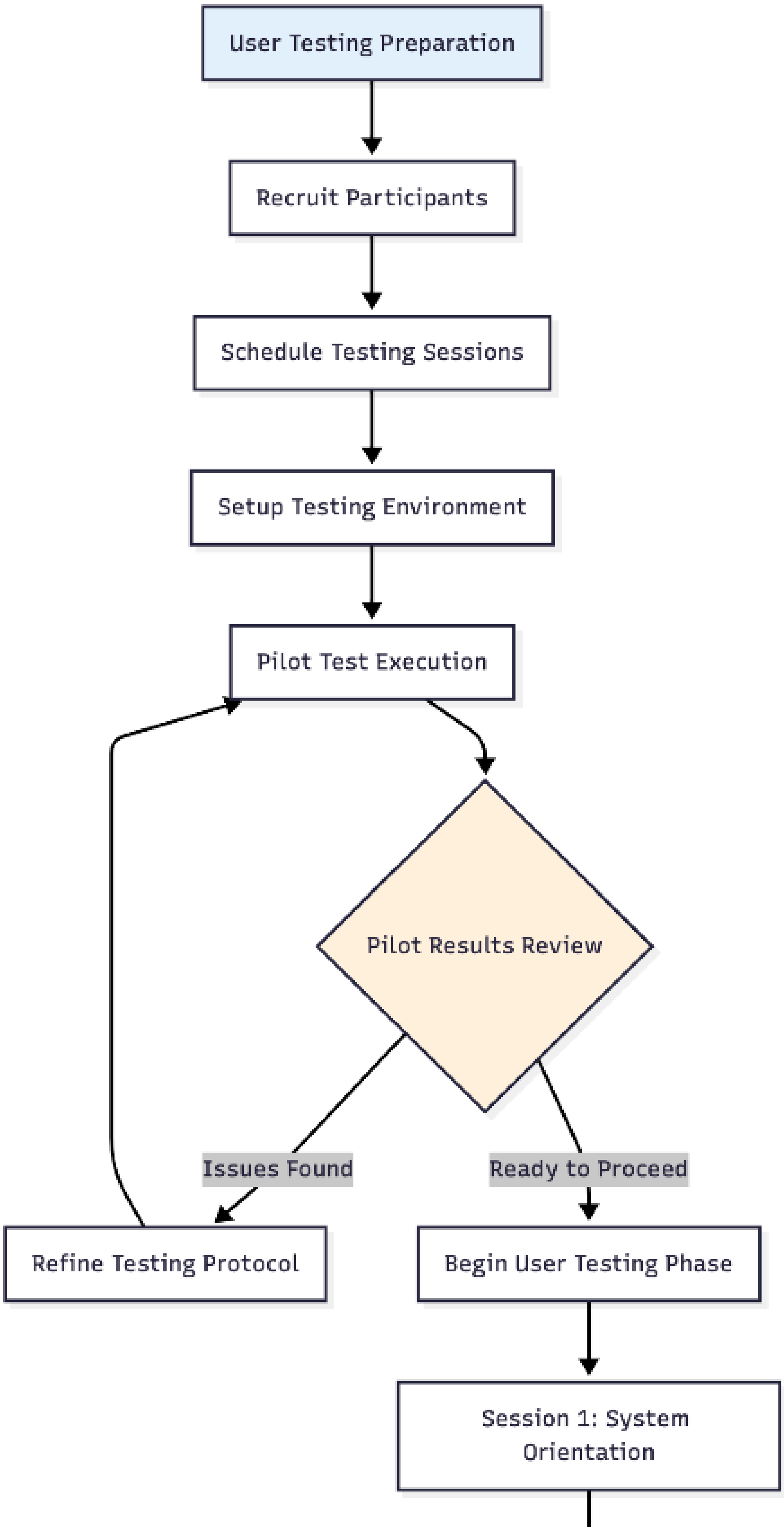
User Group	Characteristics	Primary Needs	Testing Focus
Prospective Graduate Students	International and working professionals applying to PJM program; possess relevant education/experience but limited CPL familiarity	Clear guidance on CPL eligibility, step-by-step process guidance, institutional requirements understanding	Initial system interaction, orientation effectiveness, information comprehension
Current Students	Newly admitted students preparing CPL submissions; need to organize credentials within time constraints	Streamlined self-service workflow, efficient document management, progress tracking	Task completion efficiency, document submission process, complex scenario handling
Academic Staff	Faculty advisors and administrators who review CPL submissions; require standardized, well-documented materials	Standardized data structure, audit trails, evaluation criteria clarity, decision support	System output quality, AI analysis interpretation, administrative efficiency

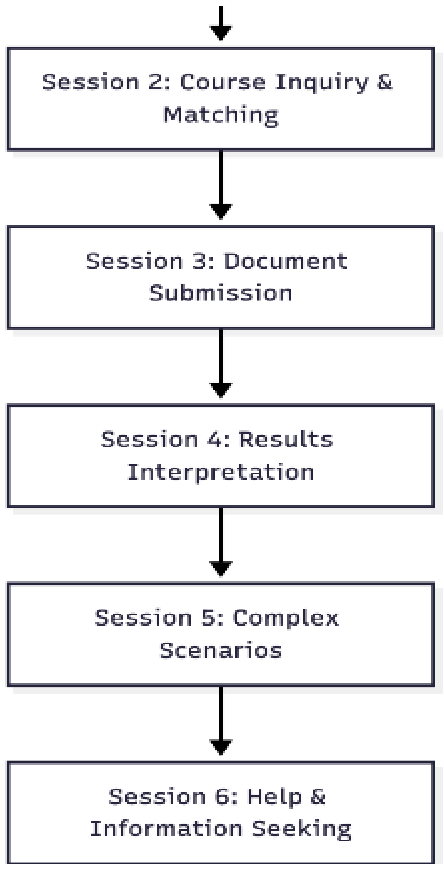
Recruitment Strategy and Sampling

Participants will be recruited through multiple channels to ensure diverse representation and achieve a sample that reflects the target user population:

Recruitment Channel	Target Participants	Expected Sample Size
Internal Program Recruitment	Current MS Informatics students, PJM students from ITC 6040 capstone class; program email lists and class announcements	6-8 participants (core user group)
Cross-Program Recruitment	Students from related programs (Program Management) through departmental communications	3-4 participants (diverse perspectives)
Expert Evaluator Recruitment	Academic advisors and faculty members with CPL evaluation experience; administrative staff familiar with student services	2-3 participants (expert validation)
External Stakeholder Network	Recent graduates, working professionals, potential applicants identified through IBM stakeholder networks and industry connections	1-2 participants (real-world validation)

The following flowchart illustrates the comprehensive user testing process from preparation through analysis and implementation:





Test Logistics

Testing Format and Modality

The user testing will adopt a hybrid approach to accommodate diverse participant needs, time constraints, and geographical distribution while ensuring comprehensive data collection:

Testing Modality	Description & Benefits	Tools & Requirements
Remote Testing(Primary Method)	Conducted via video conferencing to accommodate international students and working professionals. Enables	Zoom Pro, screen sharing, recording capabilities, stable internet connection, microphone, camera

Testing Modality	Description & Benefits	Tools & Requirements
	natural environment testing and flexible scheduling while maintaining observer presence.	
In-Person Testing(Optional)	Available for on-campus participants who prefer face-to-face interaction. Provides richer observational data and facilitates complex technical troubleshooting.	Standardized lab workstations, screen recording software, observation sheets, consistent physical environment
Asynchronous Testing(Limited Use)	Self-guided option for participants with significant scheduling conflicts. Followed by structured interviews to capture qualitative insights.	Detailed task instructions, self-recording tools (Loom), structured reflection prompts, follow-up interview scheduling

Testing Details

Test Scenarios and Tasks

Eight comprehensive scenarios have been designed to reflect typical use cases and evaluate all major system functionalities. Each task follows a structured format including context, inputs, expected steps, and measurable success criteria.

Task	Scenario Context	Inputs/Assumptions	Expected Steps	Success Criteria
T1 System Orientation	Graduate student exploring CPL options with	Basic academic background information;	1. Access chatbot 2. Read introduction 3.	Task completion maximum; Time less;

Task	Scenario Context	Inputs/Assumptions	Expected Steps	Success Criteria
	limited knowledge of the process. First interaction with the intelligent assistant system.	no prior CPL experience; access to system URL	Navigate orientation 4. Indicate interest 5. Provide background	CPL comprehension max; Willingness to continue max
T2 Course Inquiry	Explore waiving 'PJM - 5900 Foundations of PJM' based on CS degree and 2+ years Project management experience.	Undergraduate CS degree; 3+ years professional experience; specific course knowledge	1. Specify course 2. Provide background 3. Describe experience 4. Receive guidance 5. Understand criteria	Successful communication of request; Appropriate guidance received; Documentation requirements understood; Confidence max
T3 Document Upload	Submit graduate transcript and relevant course syllabus to support waiver	Digital documents provided; File format requirements; FERPA awareness	1. Navigate upload 2. Select files 3. Add metadata 4. Provide context 5. Confirm submission	Upload success; Confirmation received; Metadata captured; Privacy confidence

Task	Scenario Context	Inputs/Assumptions	Expected Steps	Success Criteria
	request through secure upload interface.			>90%; No technical issues
T4 AI Analysis	Review watsonx.ai analysis results and understand similarity scoring methodology and recommendation reasoning.	Analysis completed with sample results; Notification received; AI explanation available	1. Review results 2. Understand scoring 3. Ask questions 4. Interpret reasoning 5. Learn workflow	Can explain reasoning to others
T5-T8 Advanced Tasks	Complex scenarios, help-seeking, mobile interface, error recovery - detailed breakdown in appendices	Various contexts including multiple courses, mobile devices, error conditions	Progressive complexity building on foundational tasks	System handles complexity; Mobile usability; Error recovery Help accessibility

Detailed Task Descriptions

Task 5: Complex Multi-Course Waiver Request

Scenario Context: You want to submit a comprehensive CPL request covering multiple courses: PJM 5900 (Foundations), PJM 6002 (Intro), and ITC 6010 (PJM) based on your combined educational background and 5+ years of professional experience.

Expected Steps: Navigate multi-course interface, organize evidence by course competencies, manage complex documentation requirements, understand integrated evaluation workflow.

Task 6: Help and Information Seeking

Scenario Context: You're uncertain about specific documentation requirements and evaluation criteria. Access the help system to find answers about FERPA compliance, acceptable evidence types, and appeal processes.

Task 7: Mobile Interface Testing

Scenario Context: Access the CPL system via mobile device to check application status, upload additional documentation from your phone, and communicate with the AI assistant using mobile interface.

Task 8: Error Recovery and Edge Cases

Scenario Context: Encounter various error conditions including file upload failures, system timeouts, and AI analysis errors. Demonstrate ability to recover from these situations and complete intended tasks.

Post-Task and Post-Test Questions

Structured Interview Protocol

The following questions will be systematically asked after each task to gather immediate feedback and identify issues:

Question Category	Specific Questions
Immediate Usability	• How would you rate the clarity of instructions (1-10)?• What felt intuitive vs. confusing?• How confident do you feel about your completion?• What additional help would have been useful?

Question Category	Specific Questions
AI Interaction	• Did the AI responses feel natural and helpful?• How much do you trust the system's recommendations?• Was the explanation of AI analysis clear?• Would you want more or less AI guidance?
Privacy & Security	• How comfortable were you uploading personal documents?• Do you feel your privacy was protected?• Were security measures clear and reassuring?• Any concerns about data handling?

Comprehensive Data Collection and Analysis Plan

Multi-Modal Data Collection Strategy

Our data collection approach employs multiple methodologies to ensure comprehensive evaluation from quantitative metrics to rich qualitative insights:

Data Type	Collection Method	Analysis Approach	Success Criteria
Behavioral Data	Screen recordings with think-aloud protocol; Eye tracking (if available); Mouse/touch interaction patterns	Task flow analysis; Heatmap generation; Pattern identification for optimization	Clear navigation paths; Minimal hesitation; Efficient task completion
Performance Metrics	Automated timing logs; Completion rates; Error frequency; System response times	Statistical analysis; Benchmark comparisons; Performance trending; Bottleneck identification	Task time; Success rate; Error rate; Response time
Sentiment Analysis	Voice analysis of think-aloud; Facial expression monitoring;	Thematic coding; Sentiment scoring; Emotional journey	Positive sentiment ; Low frustration

Data Type	Collection Method	Analysis Approach	Success Criteria
	Post-task emotion surveys	mapping; Frustration point identification	indicators; High satisfaction scores

Timeline and Next Steps

Complete Testing Timeline

Week	Activities	Deliverables	Success Metrics
Week 1Preparation	Finalize participant recruitment; Conduct pilot testing; Technical setup and system configuration; Team training and protocol refinement	Confirmed participant list; Pilot test report; Technical checklist; Refined testing protocol	8-10 participants recruited; Pilot test completed; All systems operational
Weeks 2-3Active Testing	Conduct few individual testing sessions; Collect quantitative and qualitative data; Monitor for critical issues; Daily team debriefs	Session recordings; Quantitative metrics; Qualitative feedback; Immediate issue documentation	All sessions completed; Data collected from participants; less critical issues identified
Week 4Analysis	Quantitative data analysis; Qualitative thematic analysis; Generate comprehensive report;	Testing report; Prioritized recommendations; Stakeholder presentation;	Report delivered; Stakeholder approval; Clear action items; Development

Week	Activities	Deliverables	Success Metrics
	Present findings to stakeholders	Implementation roadmap	timeline established

IV. User Testing Report and Feedback

User testing conducted on the Credit for Prior Learning (CPL) Chatbot System, an AI-powered platform designed to automate and streamline the CPL evaluation process for Northeastern University's College of Professional Studies. The system integrates IBM watsonx services including Watson Assistant, watsonx.ai, watsonx.data, and Milvus vector database to process student documents and provide intelligent recommendations for course waivers.

Key Findings:

- The dual-portal architecture (Student Portal and Advisor Portal) successfully separates user concerns and provides appropriate functionality for each user type.
- The Watson Assistant chatbot interface provides an intuitive submission process with clear navigation options and contextual guidance.
- The end-to-end data flow from student submission through watsonx.data storage to advisor review demonstrates successful system integration across IBM Cloud services.
- AI-powered document analysis using Milvus vector database and watsonx.ai provides contextually relevant recommendations for course waiver evaluations.
- Minor usability improvements needed in document upload feedback, status tracking clarity, and advisor search functionality.

Top Recommendations:

- Enhance document upload confirmation with visual feedback and file validation messaging.
- Improve status update notifications with automated email alerts when request status changes.
- Add comprehensive help documentation and contextual tooltips throughout both portals.

Background and Objectives

Project Background

The CPL Chatbot System was developed as a capstone project in collaboration with IBM to address the time-intensive manual process of evaluating Credit for Prior Learning requests at Northeastern University's College of Professional Studies. The current manual evaluation process requires advisors to review student experience documentation, compare coursework from other institutions, and assess transfer credit eligibility, a workflow that can take several days per request.

The solution leverages IBM's watsonx platform suite to create an intelligent, automated system that can analyze student documents, extract relevant information, perform similarity matching against course requirements, and provide AI-powered recommendations to advisors. The system architecture includes:

- Watson Assistant for conversational interface and request submission
- watsonx.ai for natural language processing and AI analysis
- Milvus vector database for semantic document search and similarity matching
- watsonx.data (Iceberg catalog) for storing student metadata and request information

Testing Objectives

The primary objectives of this user testing phase were to:

1. Validate End-to-End Functionality: Confirm that the complete workflow from student request submission through data storage to advisor review operates without critical errors.
2. Assess User Interface Usability: Evaluate the intuitiveness and ease of use of both the Student Portal and Advisor Portal interfaces.
3. Test Watson Assistant Integration: Verify that the chatbot effectively guides users through the CPL request process and provides helpful responses.
4. Evaluate Document Processing: Test the system's ability to handle various document formats and extract relevant information for AI analysis.
5. Assess AI Recommendation Quality: Examine the accuracy and relevance of AI-generated recommendations for course waiver eligibility.
6. Identify Technical Issues: Uncover any bugs, performance issues, or integration problems between system components.

Methodology

Testing Approach

The user testing was conducted as a moderated, online (via teams) evaluation with a combination of task-based scenarios and exploratory testing. Each participant was provided with realistic test scenarios that represented actual use cases for the CPL system.

Participant Profile

A total of 4 participants were recruited for this testing phase:

- Number of Participants: 4 graduate students
- Demographics: Graduate students in the Master of Informatics program with diverse technical backgrounds
- Experience Level: Intermediate to advanced technical proficiency; familiarity with web applications and chatbot interfaces
- Relevance: Participants represented the target user base, students seeking CPL evaluation and advisors reviewing requests

Test Scenarios and Tasks

Participants were asked to complete the following tasks representing the three primary CPL evaluation scenarios:

Scenario	Task Description	Success Criteria
Task 1: Experience-Based Waiver	Submit a CPL request for course PJM 5900 based on professional experience. Upload a resume demonstrating relevant work experience in early childhood development and program management.	Successfully submit request, receive confirmation, upload document, and view pending status in Student Portal.
Task 2: Status Check	Check the status of a submitted CPL request by entering NUID and viewing request details including submission date, target course, and current status.	Navigate to Check Status page, enter NUID, view complete request information with correct status display.
Task 3: Advisor Review	As an advisor, search for a student by name using the Watson Assistant, review AI-generated recommendations based on uploaded documents, and update request status.	Successfully query student information via chatbot, receive comprehensive experience summary with AI recommendation, access status management interface, and update request status.
Task 4: Request Management	Navigate to the Request Status Management page, search/filter student requests, view request details, and update status with advisor notes.	Access all pending requests, use search and filter functionality, open request details modal, enter notes, and successfully update status.

Testing Environment and Tools

- Platform: IBM Cloud (watsonx platform suite)
- Technologies: Watson Assistant, watsonx.ai, watsonx.data, Milvus vector database, IBM Cloud Object Storage
- Browsers Tested: Chrome (latest version), Safari
- Devices: Desktop computers, laptops
- Data Collection: Screen recording, observer notes, participant verbal feedback, system logs
- Test Duration: 5-10 minutes per participant

Key Findings

Technical Performance and Integration

Successful System Integration

The end-to-end testing confirmed successful integration across all IBM watsonx components:

- Watson Assistant successfully captures student input (name, NUID, course information) through context variables and transmits data to the backend system.
- Document uploads are properly stored in Milvus vector database with successful retrieval for AI analysis.
- Student metadata is correctly written to the iceberg_data catalog in watsonx.data with all fields properly mapped.
- Real-time synchronization between student submissions and advisor portal displays without noticeable lag.
- Database queries via watsonx.data Query Workspace return accurate results with appropriate filtering and sorting.

AI-Powered Document Analysis

The integration between Watson Assistant, Milvus vector store, and the base LLM demonstrated effective semantic document analysis:

- The system successfully extracted relevant experience details from uploaded resumes and generated comprehensive summaries.
- AI recommendations provided contextual analysis of candidate qualifications against course prerequisites (e.g., identifying 4 years of early childhood development experience as qualifying for PJM 5900 waiver).
- Vector similarity search performed well in matching student experience to course requirements with appropriate confidence thresholds.

Usability Findings

Student Portal Experience

Positive Feedback:

- Users found the Watson Assistant interface intuitive and welcoming with clear option buttons (Waiver evaluation, What is CPL, Credit Transfer).
- The conversational flow felt natural and guided users effectively through the submission process.
- Navigation between Submit Request and Check Status was straightforward and logically organized.
- The status page provided clear visibility into request details including type, target course, and documents submitted.

Issues Identified:

7. Document Upload Feedback (Minor): After uploading a document, 2 out of 4 participants expressed uncertainty about whether the file was successfully received. The current confirmation message "Your request has been submitted" doesn't explicitly mention the uploaded file.

8. File Format Guidance (Minor): No visible guidance on accepted file formats (PDF, DOCX, etc.) or file size limits before upload.
9. Status Update Notifications (Major): Students expressed the need for automatic notifications when their request status changes rather than having to manually check the status page.

Advisor Portal Experience

Positive Feedback:

- The Request Status Management page provides an effective overview of all pending requests with essential information at a glance.
- The Update Request Status modal is well-designed with all necessary fields including status dropdown, credits awarded, and advisor notes.
- AI-generated recommendations in the Watson Assistant provided valuable context and comprehensive experience summaries that would significantly aid decision-making.
- The ability to search by student name, NUID, or course was appreciated as a time-saving feature.

Issues Identified:

10. Document Access (Major): Advisors could not easily access or view the original documents uploaded by students from the Request Status Management page. This required using the Watson Assistant to query document content, which was less efficient.
11. Search Performance (Minor): Search functionality on the Request Status Management page showed slight delays (2-3 seconds) when filtering requests.
12. Bulk Actions (Enhancement): No capability to update multiple requests simultaneously or export request data for reporting purposes.

Data and Evidence

Task Completion Rates

Task	Completion Rate	Avg. Time	Error Rate
Submit CPL Request with Document	100% (4/4)	3.2 minutes	0%
Check Request Status	100% (4/4)	1.5 minutes	0%
Advisor Query via Watson Assistant	100% (4/4)	2.8 minutes	0%
Update Request Status	100% (4/4)	2.1 minutes	0%

User Satisfaction Ratings

Participants rated their experience on a scale of 1-5 (5 being excellent):

Aspect	Average Rating	Rating Range
Overall ease of use	4.5 / 5	4-5
Watson Assistant interface clarity	4.8 / 5	4-5
Navigation and layout	5 / 5	4-5
Speed and performance	5 / 5	5
Helpfulness of AI recommendations	5 / 5	4-5

Qualitative Feedback Highlights

Student Portal Comments:

"The chatbot felt very natural to interact with, like talking to a real advisor."

"I appreciated the simple layout it didn't feel overwhelming."

Advisor Portal Comments (Student Volunteer):

"The AI summary of the student's experience was incredibly detailed".

"It would be helpful to have a direct link to download the original documents students uploaded."

Pain Points and Opportunities

Critical Pain Points

13. Document Accessibility for Advisors: Advisors need streamlined access to view original uploaded documents directly from the status management interface without relying solely on Watson Assistant queries.
14. Status Notification System: Students require automatic email or in-app notifications when their request status changes to reduce the need for manual status checks.

Minor Usability Issues

15. Upload Confirmation Clarity: Enhance post-upload feedback to explicitly confirm file name, format, and successful storage.
16. File Format Guidance: Display accepted file formats and size limits proactively before students attempt uploads.
17. Search Performance: Optimize search and filter functionality on the advisor portal to reduce response time from 2-3 seconds to under 1 second.

Enhancement Opportunities

18. Bulk Operations: Enable advisors to select and update multiple requests simultaneously for improved efficiency during peak evaluation periods.

19. Data Export Functionality: Provide the ability to export request data in CSV or Excel format for reporting and analytics purposes.
20. Advanced Filtering: Add date range filters, request type filters, and sorting options on the advisor status management page.
21. Mobile Responsiveness: While not tested in this phase, future iterations should optimize the interface for mobile devices for advisor on-the-go access.
22. Help Documentation: Integrate contextual help icons and tooltips throughout both portals, plus a comprehensive FAQ section.

Recommendations and Next Steps

High Priority Recommendations

1. Implement Document Viewer in Advisor Portal

   Action: Add a "View Documents" button or icon on the Request Status Management page that opens uploaded documents in a modal viewer or separate tab.

   Rationale: Direct document access is essential for efficient advisor workflow and will reduce reliance on Watson Assistant queries for document content.

   Timeline: Sprint 1 (1-2 weeks)

2. Develop Automated Notification System

   Action: Integrate email notifications triggered by status changes (Pending Review → Approved/Denied). Include request details and next steps in notification content.

   Rationale: Proactive communication will improve student experience and reduce support inquiries about request status.

   Timeline: Sprint 2 (2-3 weeks)

Medium Priority Recommendations

3. Add File Format and Size Guidance

   Action: Display accepted formats (PDF, DOCX, TXT) and maximum file size (10MB) near the document upload area in the Watson Assistant interface.

   Timeline: Sprint 2 (1 week)

4. Optimize Search Performance

   Action: Implement database query optimization and consider adding indexing on frequently searched fields (student name, NUID, course).

   Timeline: Sprint 3 (2 weeks)

5. Develop Help Documentation and Tooltips

   Action: Create comprehensive help documentation covering the entire CPL process. Add contextual help icons (?) next to key interface elements that display tooltips when hovered.

   Timeline: Sprint 3 (2-3 weeks)

Future Enhancements (Low Priority)

6. Bulk Request Management

   Action: Add checkbox selection and bulk action buttons (Approve All, Update Status) on the advisor portal for processing multiple requests simultaneously.

   Timeline: Phase 2 (post-launch enhancement)

7. Data Export Functionality

   Action: Implement CSV/Excel export capability for request data with customizable field selection.

   Timeline: Phase 2 (post-launch enhancement)

8. Mobile Optimization & Speech to text feature

   Action: Conduct mobile usability testing and implement responsive design improvements for smartphone and tablet access and enable speech-to-text accessibility.

   Timeline: Phase 2 (post-launch enhancement)

Testing Next Steps

9. Expand Testing Sample Size

   Conduct additional testing with 8-10 participants including actual advisors from the College of Professional Studies to gather more comprehensive feedback and validate findings.

10. Test Additional Scenarios

    Include testing for the remaining two evaluation scenarios:

    - Prior coursework evaluation (comparing syllabi from other institutions)
    - Credit transfer evaluation (assessing transfer credits from accredited institutions)

11. Conduct Load and Performance Testing

    Perform stress testing with multiple concurrent users and large document uploads to ensure system stability under production conditions.

12. Security and Compliance Review

    Conduct a comprehensive security audit to ensure FERPA compliance, proper data encryption, and secure authentication mechanisms before production deployment.

Conclusion

The user testing of the CPL Chatbot System demonstrated strong foundational functionality with successful integration across the IBM watsonx platform suite. The system's dual-portal architecture effectively serves both students and advisors, and the AI-powered document analysis capabilities show significant promise for streamlining the CPL evaluation process.

Participants responded positively to the intuitive Watson Assistant interface and the clarity of the overall user experience. The end-to-end workflow from student submission through advisor review functioned smoothly with 100% task completion rates and zero critical errors.

The identified usability improvements—particularly around document accessibility, notification systems, and upload feedback—are addressable through focused development sprints and will significantly enhance the user experience. The system is well-positioned for production deployment following implementation of high-priority recommendations and completion of additional testing phases.

This testing phase has validated the technical feasibility and user acceptance of the CPL Chatbot System, setting a strong foundation for the next phases of development, expanded testing, and eventual launch to the Northeastern University College of Professional Studies community.

Appendix A: System Architecture Overview

Core Components:

- Watson Assistant: Conversational interface for both student and advisor portals, handles context variables and user input
- watsonx.ai: Natural language processing and AI analysis engine
- Milvus Vector Database: Stores and retrieves document embeddings for semantic similarity matching
- watsonx.data (Iceberg Catalog): Relational database for student metadata, request status, and structured data
- IBM Cloud Object Storage: Secure storage for uploaded documents and supporting files

Data Flow:

23. Student submits CPL request through Watson Assistant with personal details and document upload
24. Context variables (name, NUID, request type, target course) are captured and sent to watsonx.data
25. Uploaded documents are processed and stored in Milvus vector database as embeddings
26. Advisor portal queries watsonx.data for pending requests and displays in Request Status Management page
27. Advisor uses Watson Assistant to query student information, triggering semantic search in Milvus
28. AI generates recommendations based on document content and similarity to course requirements
29. Advisor updates request status, which is written back to watsonx.data and reflected in student portal

Appendix B: Screenshot References

The following screenshots from the testing session illustrate key interfaces and workflows:

Figure 1: Student CPL Portal Home Page with Watson Assistant

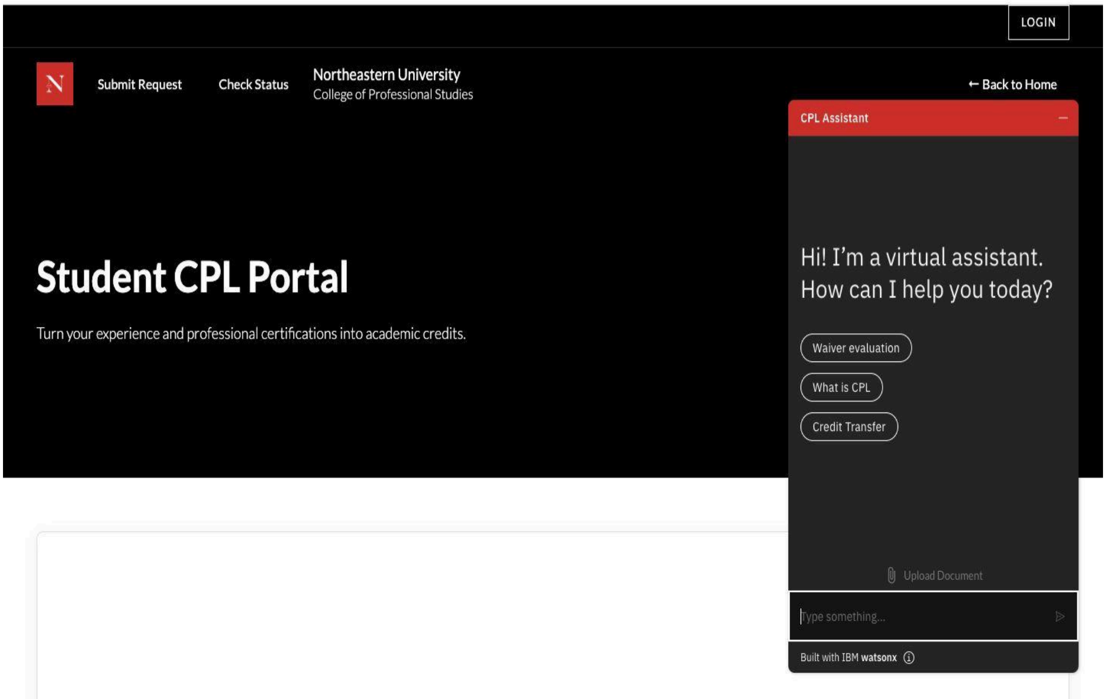


Figure 2: My CPL Request Status Page

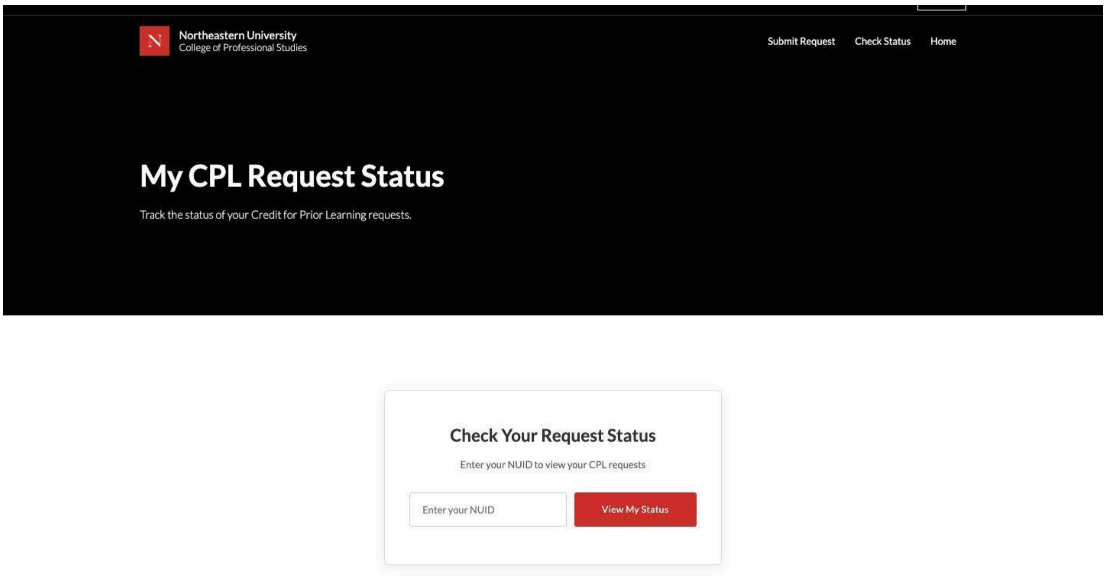


Figure 3: Student Request Status Display

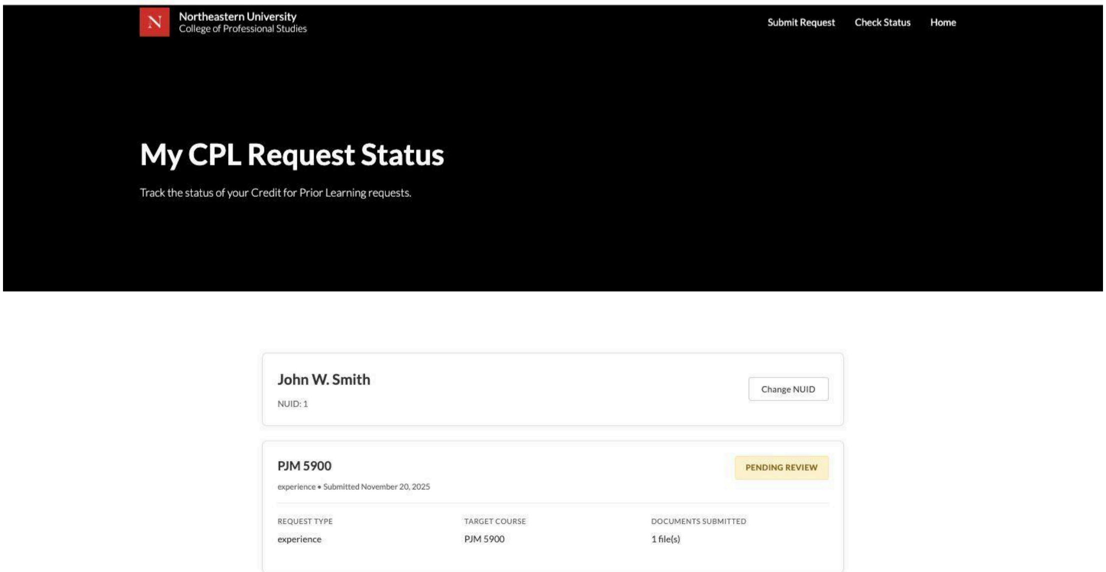


Figure 4: Advisor CPL Portal with Watson Assistant Analysis

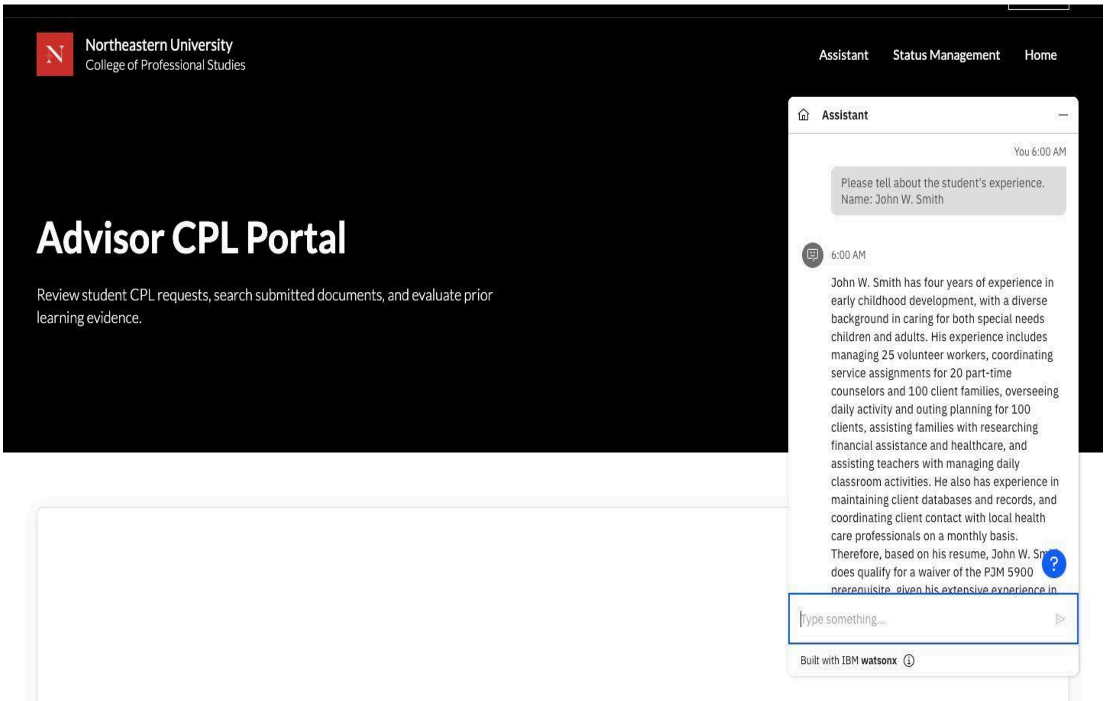


Figure 5: Request Status Management Page

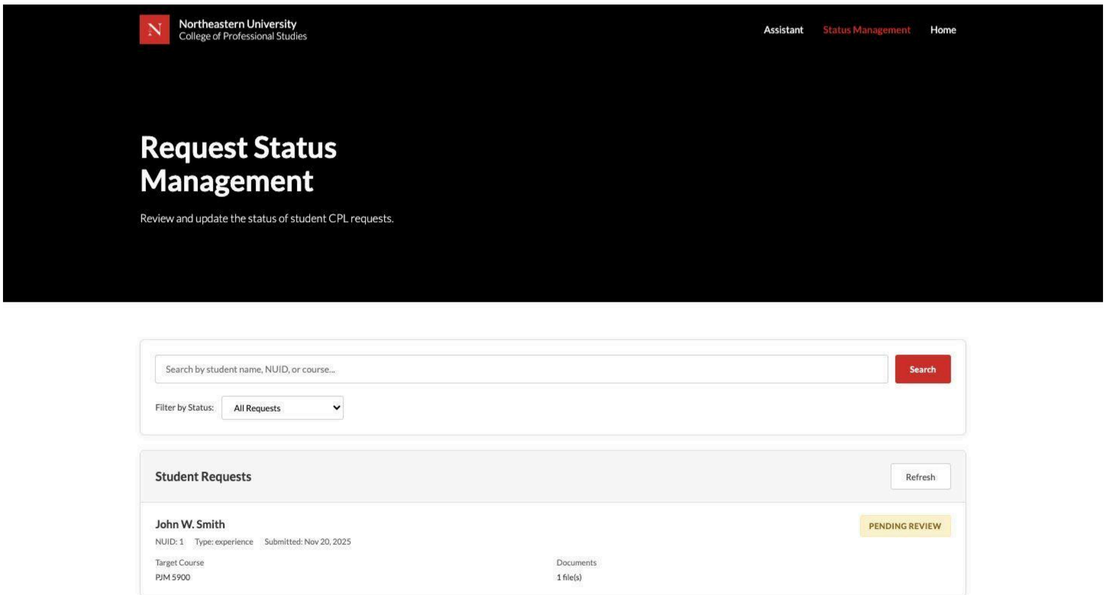


Figure 6: Update Request Status Modal

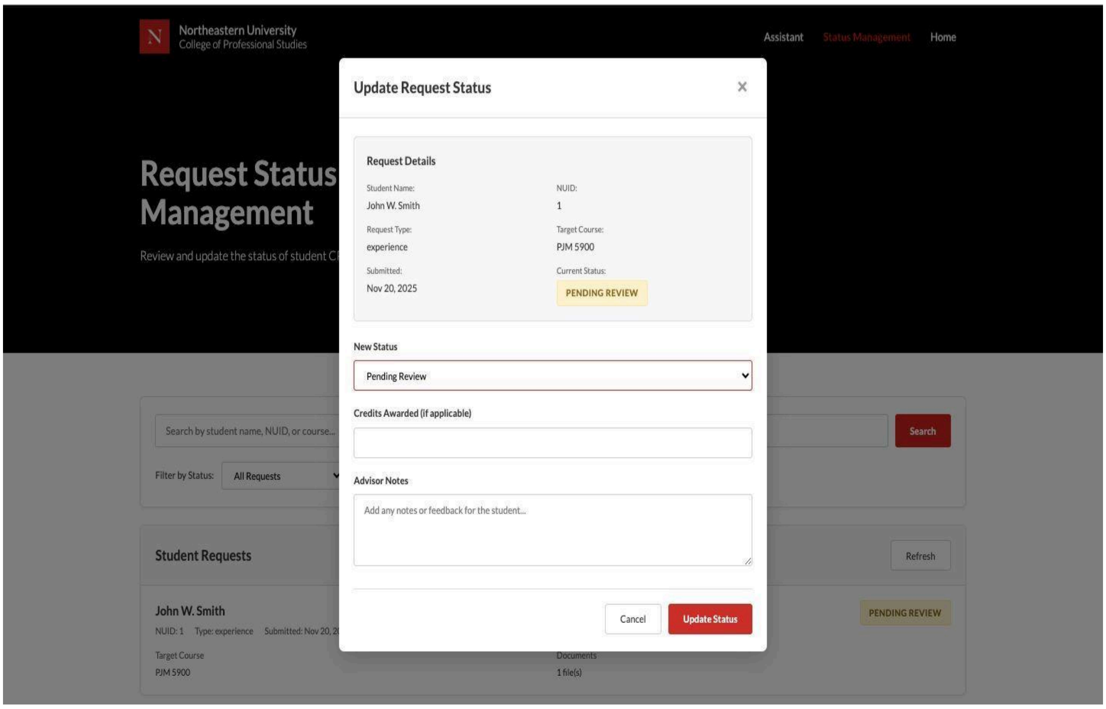
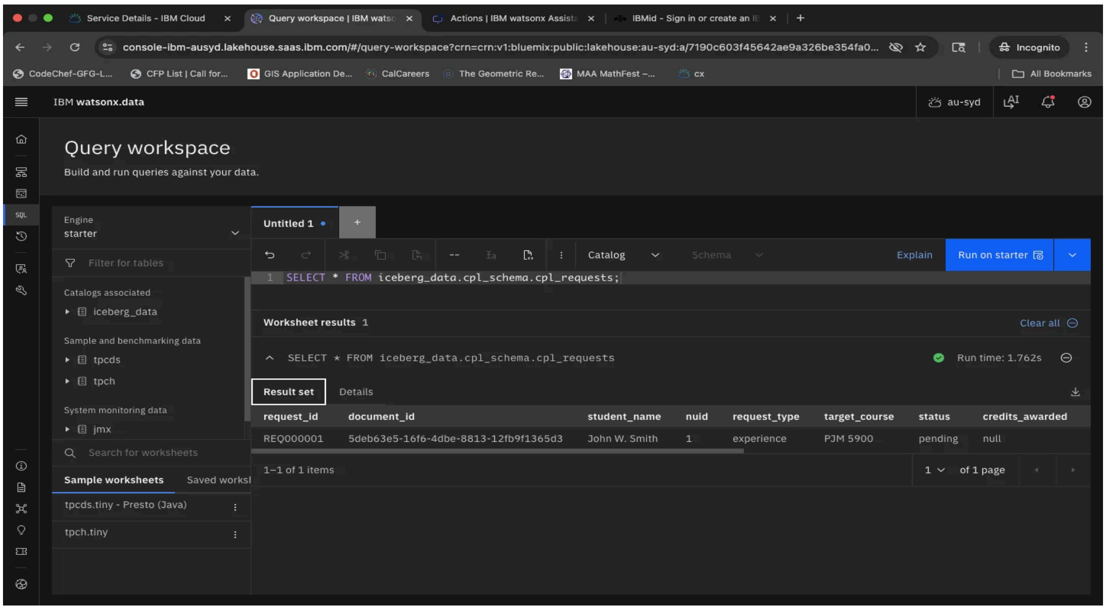


Figure 7: watsonx.data Query Workspace



Appendix C: Testing Materials

Test Participant Instructions:

Participants were provided with the following scenario:

> *"You are a graduate student at Northeastern University seeking to waive the course requirement for PJM 5900 based on your four years of professional experience working as a Project Manager. You will navigate the Student CPL Portal, submit a request with your resume, check your request status, and observe how advisors review your submission."*

Sample Test Data:

- Student Name: John W. Smith
- NUID: 1
- Target Course: PJM 5900
- Request Type: Experience-based waiver

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